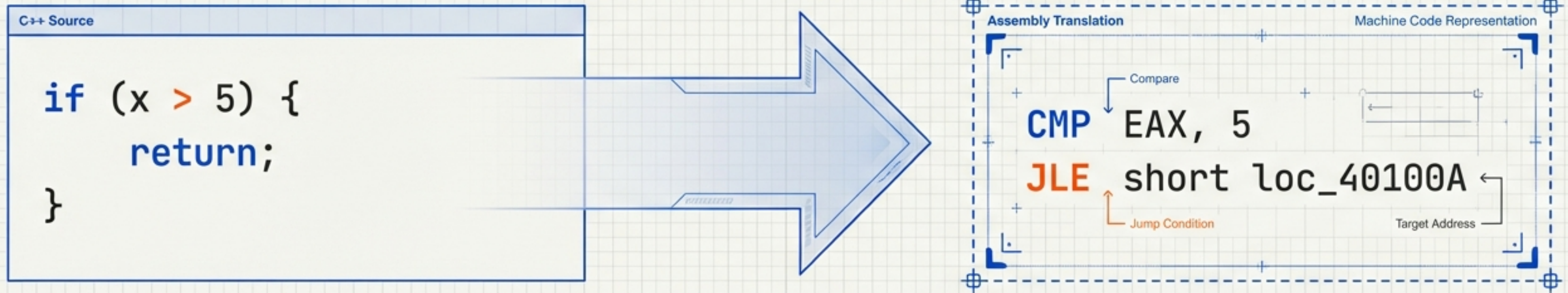


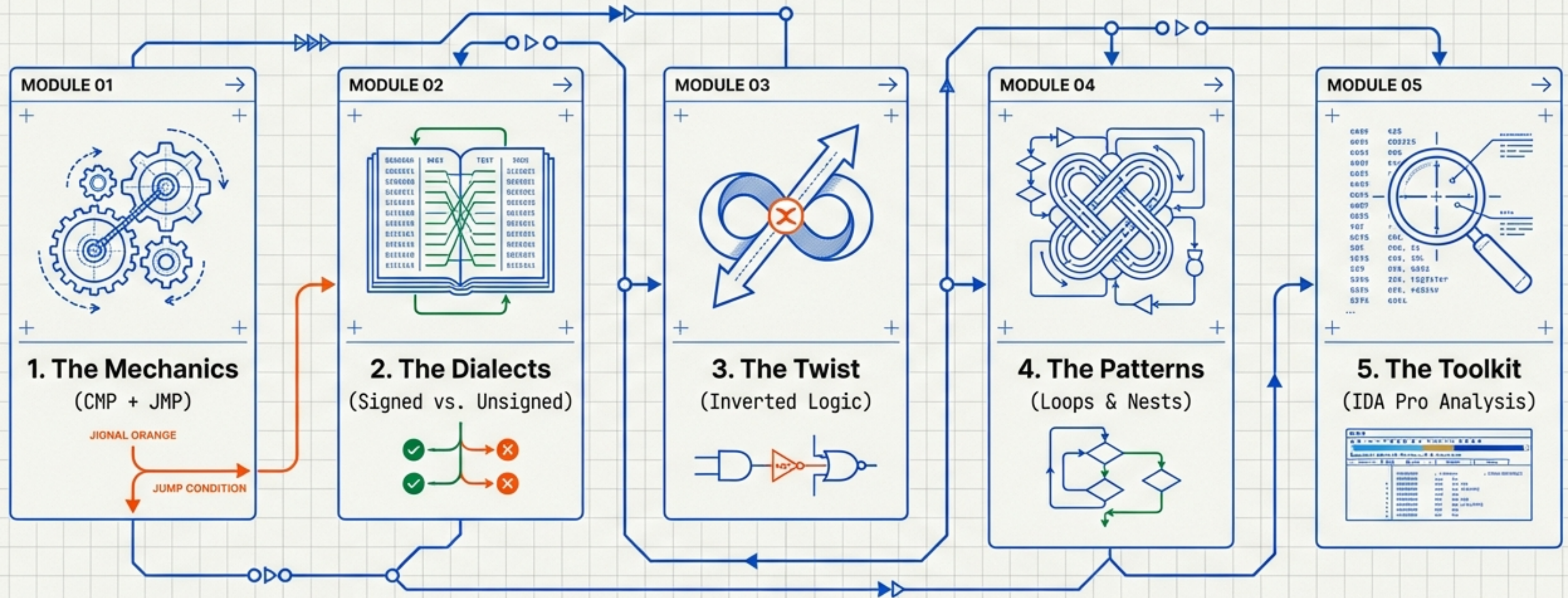
Code Constructs: The Anatomy of IF Statements

Mechanics, Logic, and Analysis in Assembly.

Translation

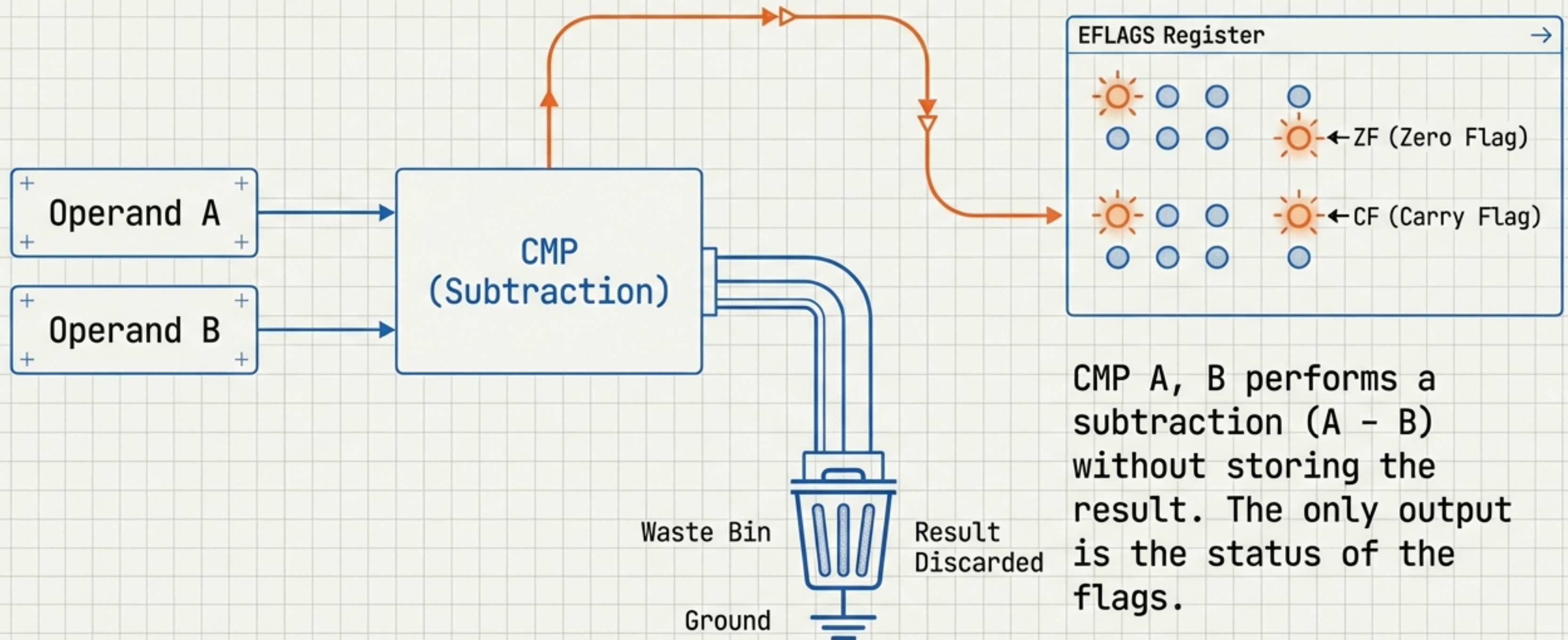


The Decoder's Syllabus



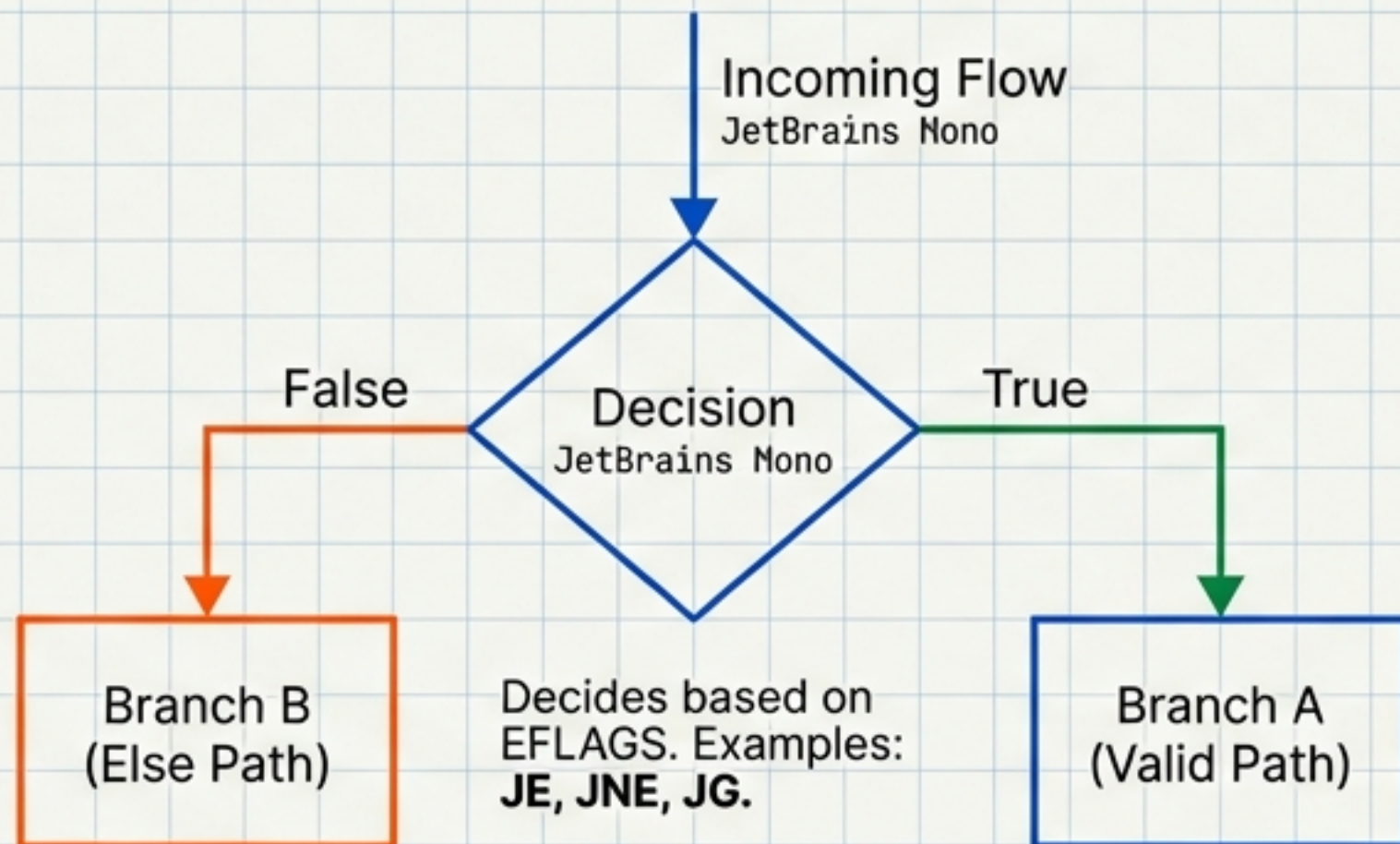
Objective: Move from atomic instructions to investigative analysis.

The Silent Mechanics of Comparison (CMP)

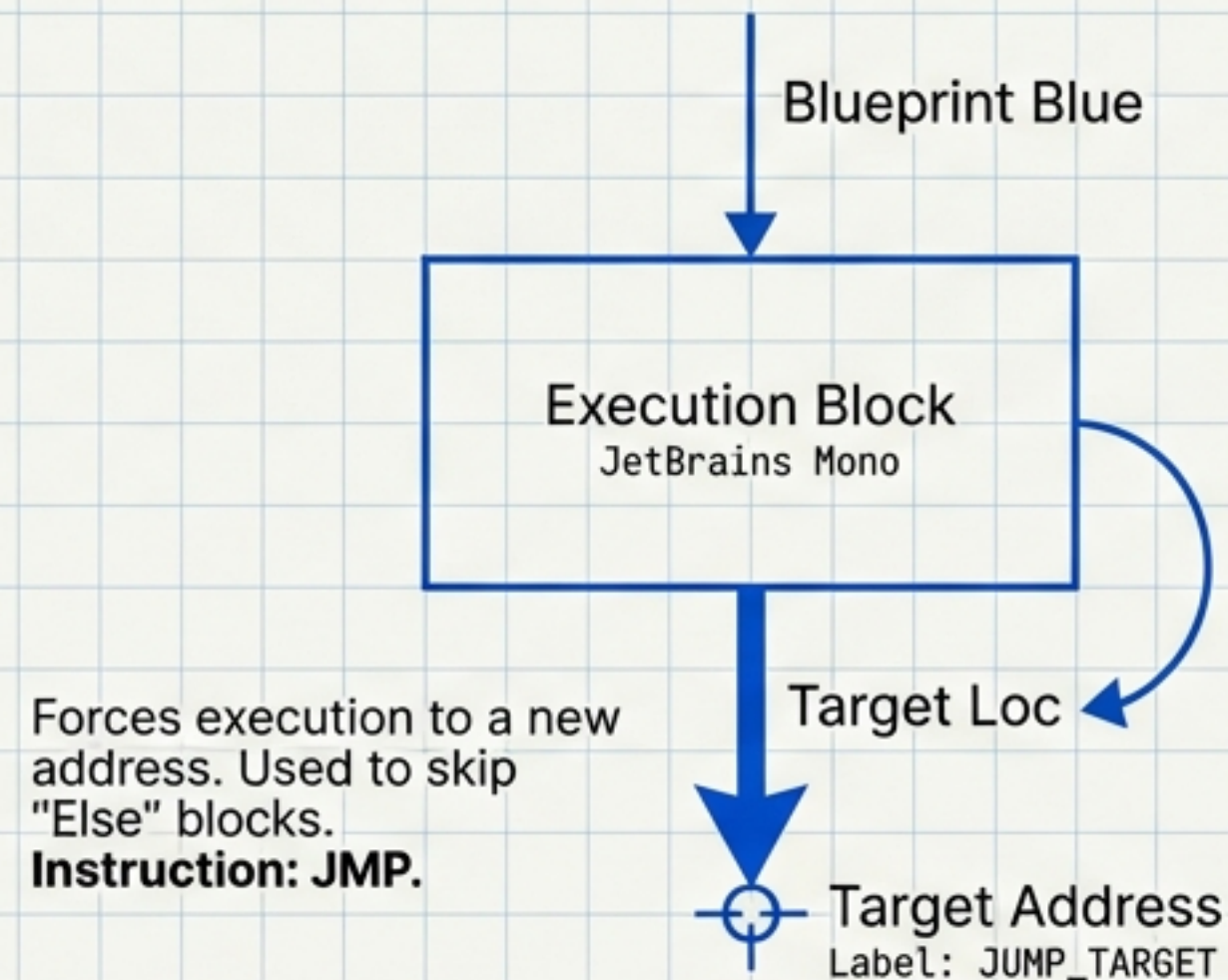


The Pivot Point: Jumps

Conditional Jump (The "If")



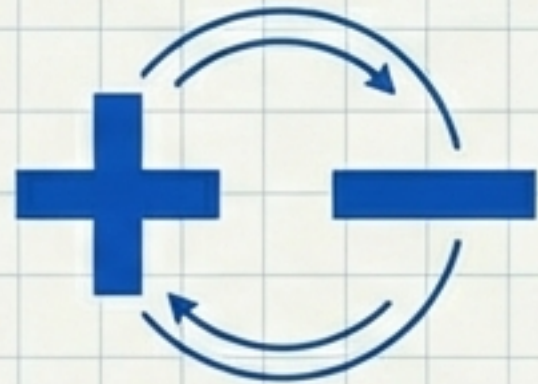
Unconditional Jump (The "Skip")



Context Clues: Signed vs. Unsigned

SIGNED DATA

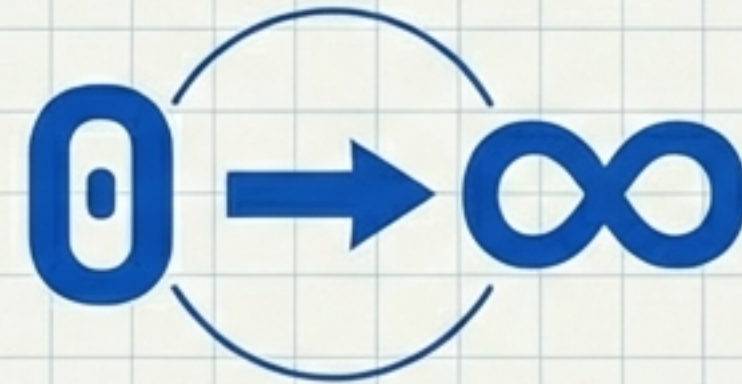
Integers (+/-)



JL (Jump Less)
JG (Jump Greater)
JLE (Jump Less Equal)
JGE (Jump Greater Equal)

UNSIGNED DATA

Positive Only (Addresses/Sizes)



JB (Jump Below)
JA (Jump Above)

The instruction defines the type. Assembly does not use 'int' or 'uint'.

The Mnemonic Device

Memorize this rule to instantly parse data types.

GL

Greater / Less

For Signed Integers

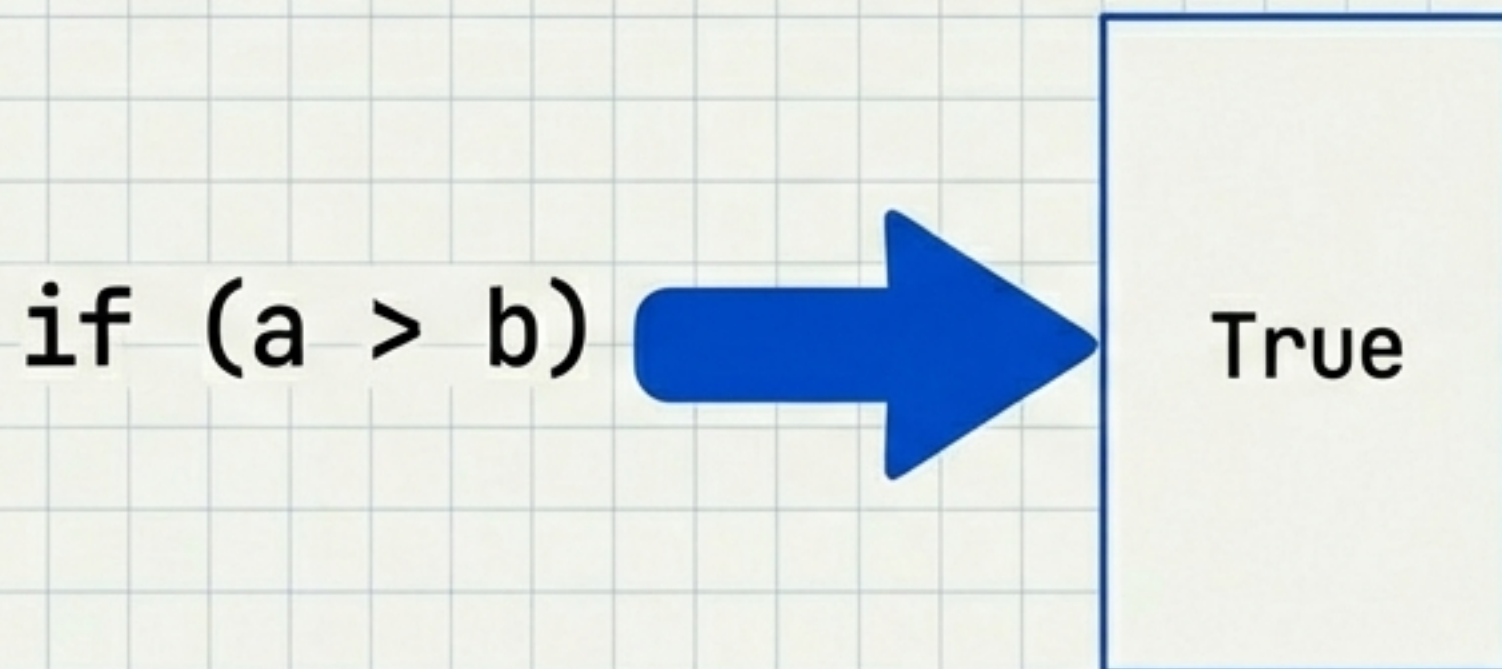
AB

Above / Below

For Unsigned Memory

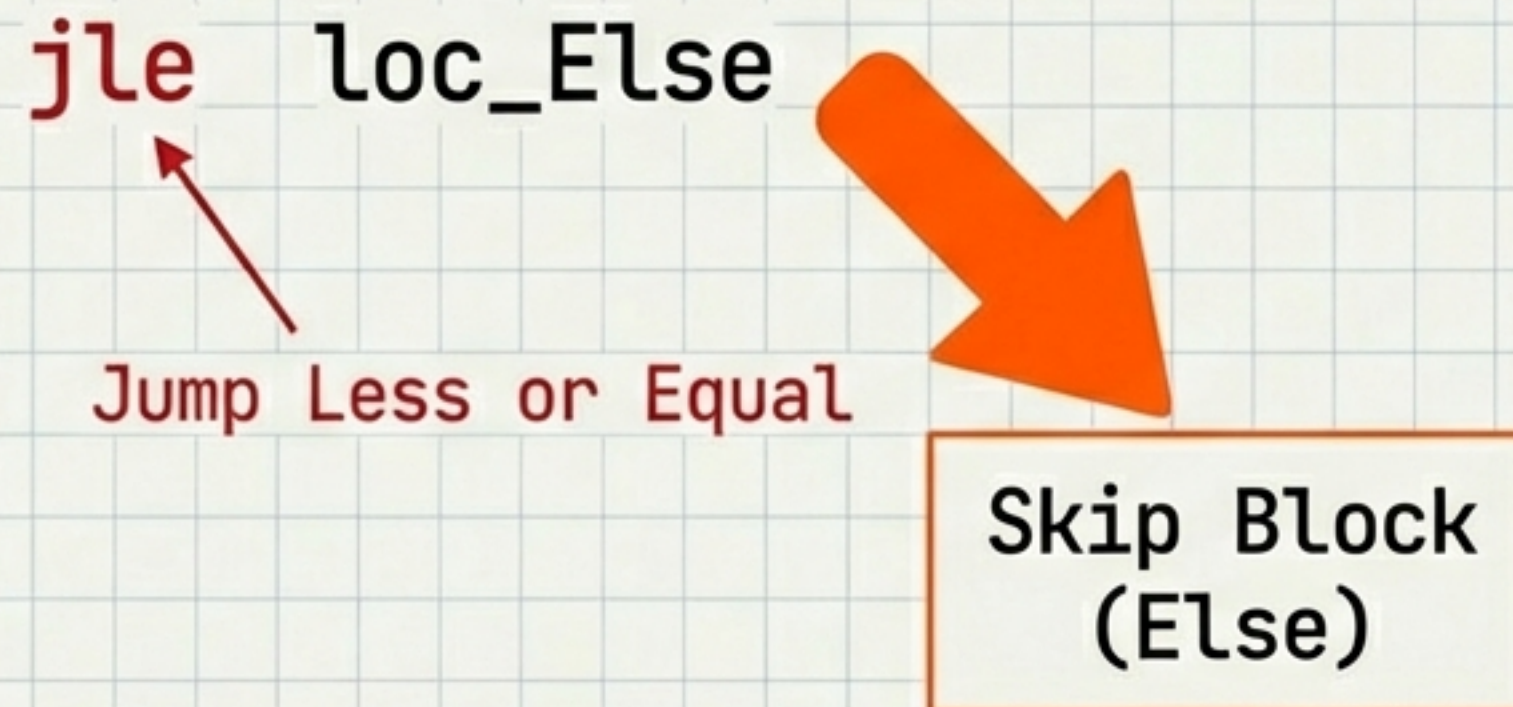
The Logic Trap: Expectation vs. Reality

Expectation (C++)



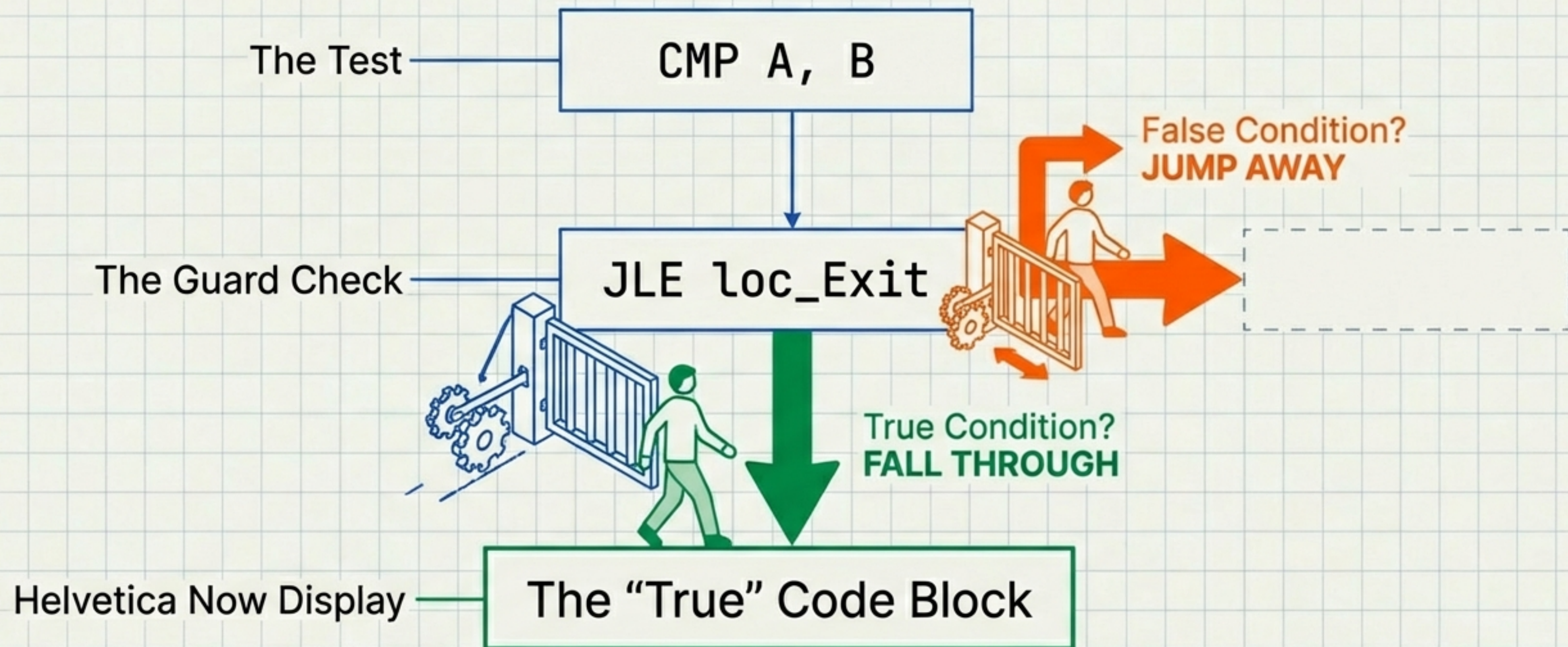
We logic for the positive.

Reality (ASM)

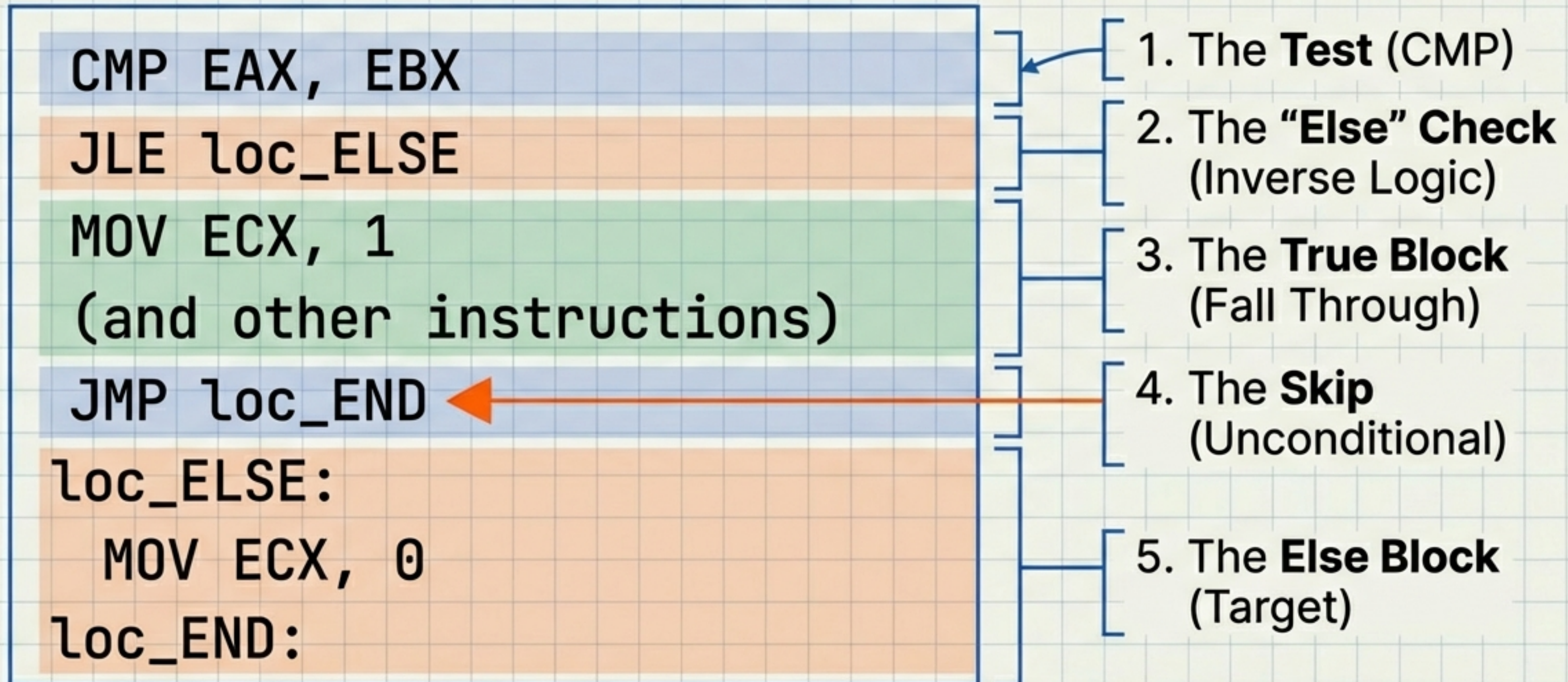


The compiler checks for the FALSE condition to skip the block.

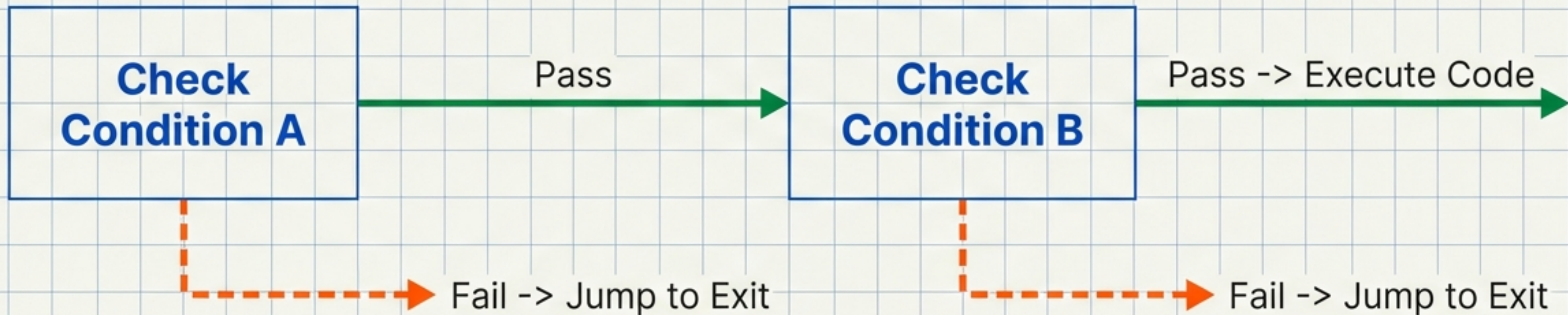
Deconstructing the “Fall Through”



Anatomy of an If/Else Block



Complex Constructs: Short - Circuiting

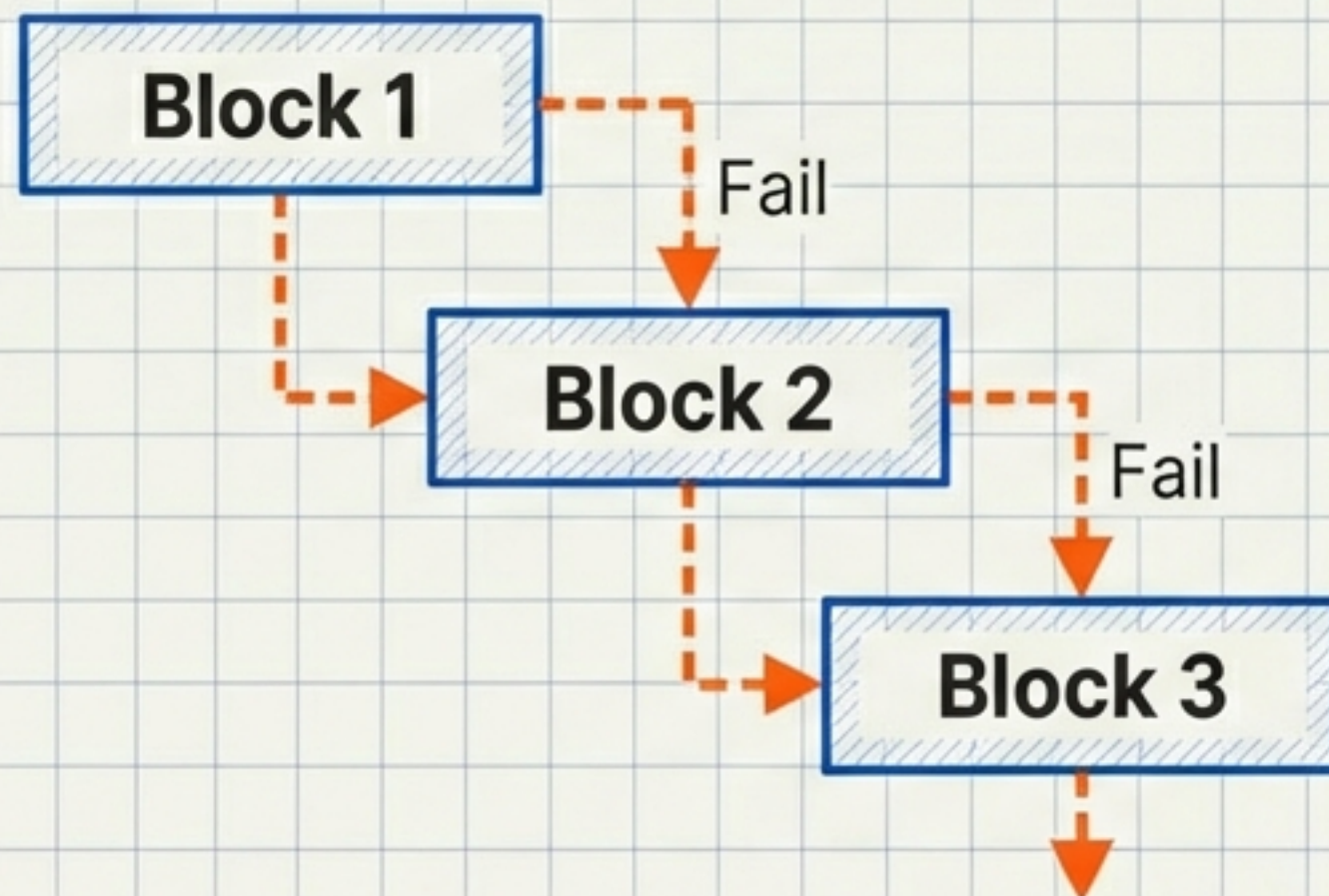


JetBrains Mono Regular in Charcoal (#222222)

Efficiency Rule: If the first condition fails, the second is never touched.

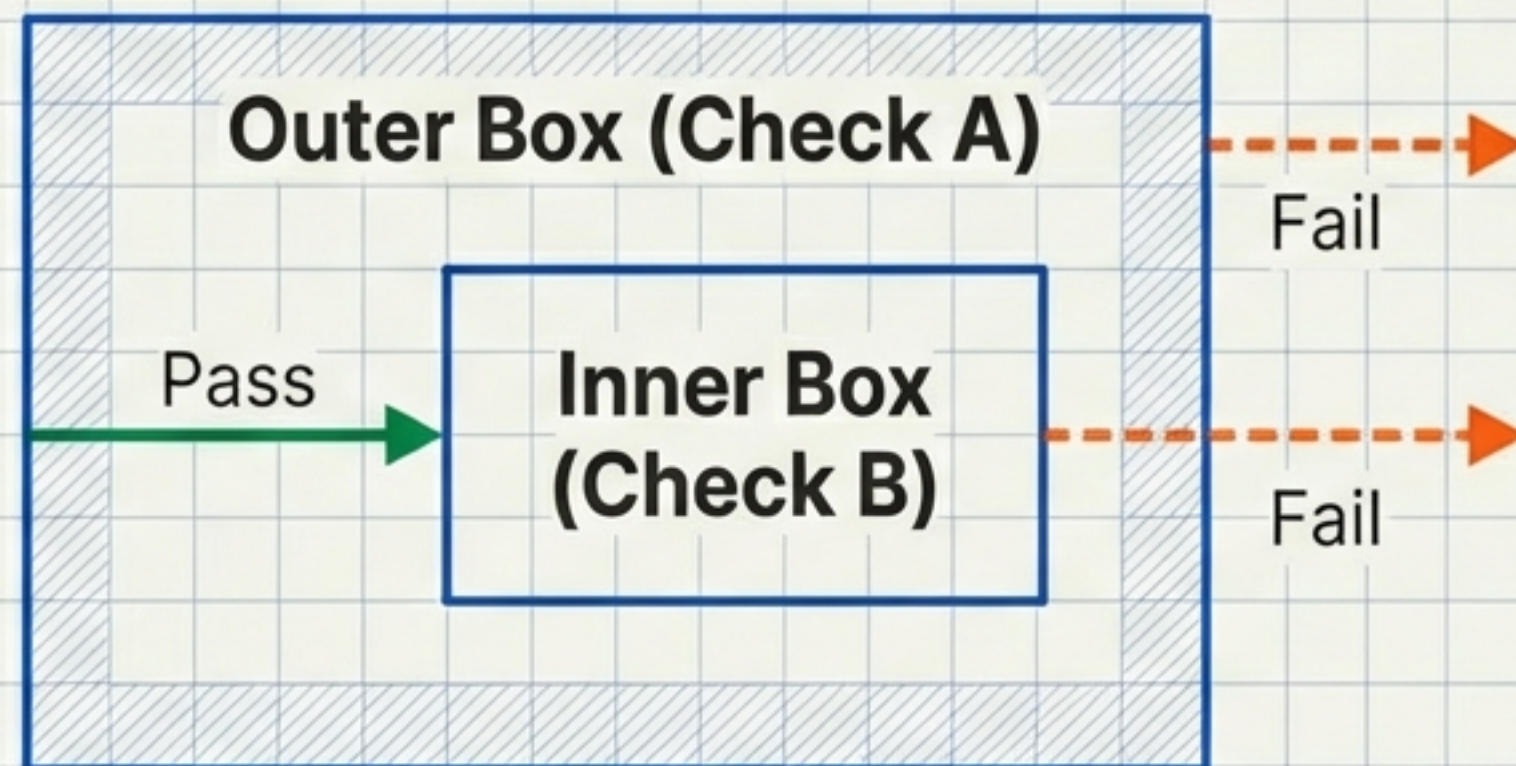
Nesting and Chains

Else-If Chain



JetBrains Mono (Regular)
Sequential checks. Multiple
CMP/JMP pairs.

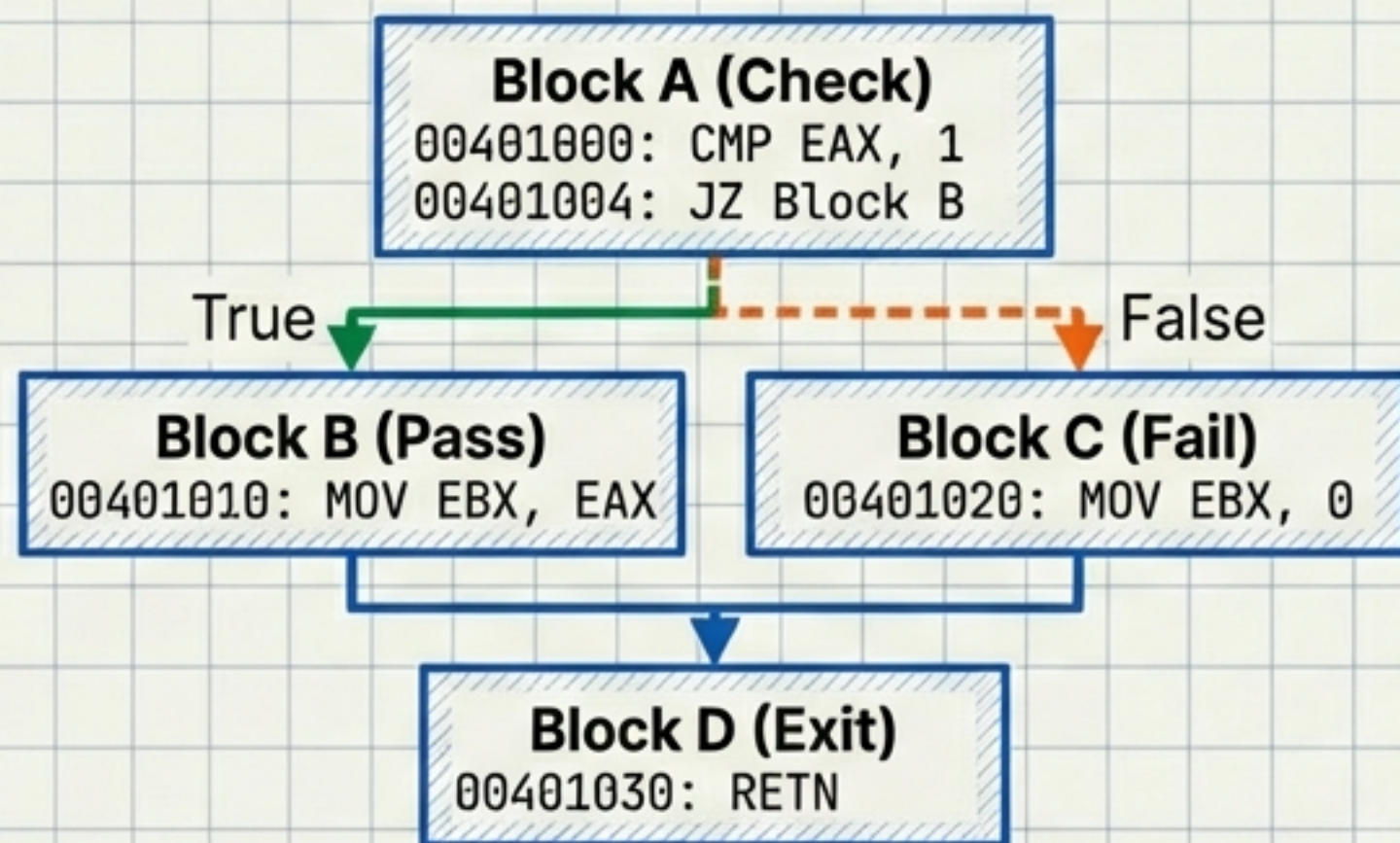
Nested If



JetBrains Mono (Regular)
Hierarchical dependency. Inner check
is only reachable if Outer passes.

The Toolkit: Graph vs. Text View

Graph View



JetBrains Mono (Regular)
Visualizing Flow & Shapes.

Text View

```
00401000: CMP EAX, 1
00401004: JZ  00401010 (Block B)
00401006: MOV EBX, 0
0040100B: JMP 00401030 (Block D)
00401010: MOV EBX, EAX
00401015: JMP 00401030 (Block D)
...
00401030: RETN
```

JetBrains Mono (Regular)
Verifying Linear Instruction Order.

Use Graph to see the forest. Use Text to read the trees.

Analysis Technique: Back-Tracing

```
00401020: MOV  EBX, [EBP+8]
00401023: PUSH EBX
00401025: CALL GetUserLevel; EAX stores result
0040102A: MOV  [EBP-4], EAX
0040102D: ...
00401032: MOV  EDX, [EBP-4]
00401035: MOV  EAX, EDX
00401037: CMP  EAX, 5
0040103A: JGE  Block_GrantAccess
```

Trace the data source. To understand the comparison, find where the register was last written.

Use Back-Tracing to understand data flow and register origins.

Analysis Technique: Renaming

Before and After

Generic Labels (Chaos)

```
CMP var_4, 10  
JLE short loc_40200
```

Renaming variables and locations progressively clarifies the code logic.

Meaningful Names (Order)

```
CMP user_attempts, 10  
JLE short loc_LoginFail
```

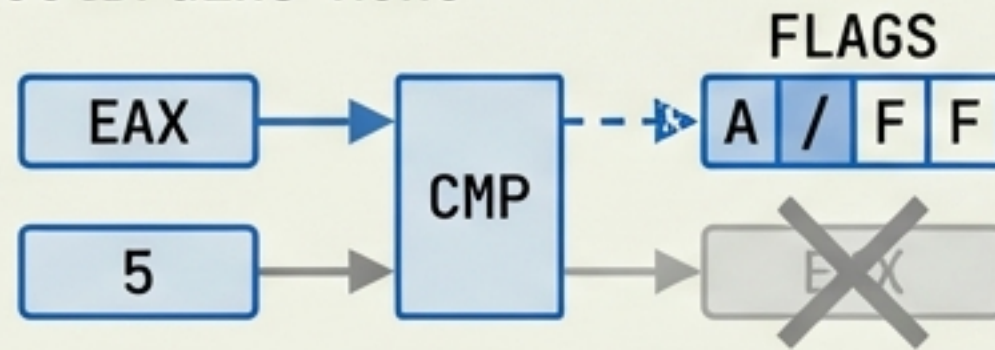
Renaming variables and locations progressively clarifies the code logic.

The Decoder's Cheat Sheet

Quick reference for assembly code analysis and reverse engineering.

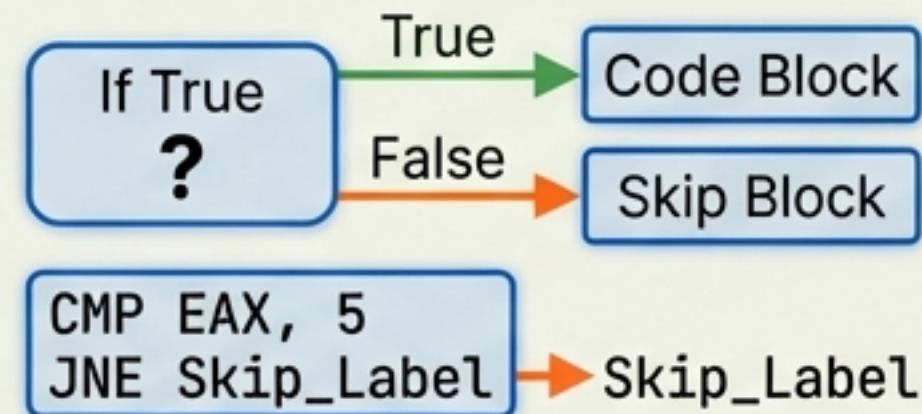
Silent CMP

JetBrains Mono



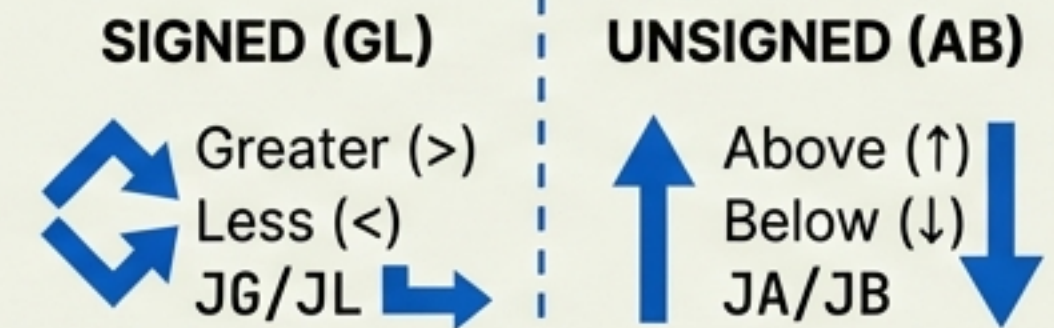
Comparison sets flags only.
No data storage.

Inverted Logic



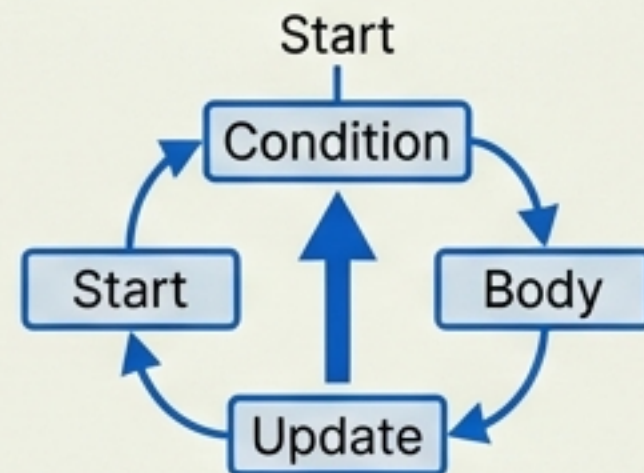
Expect the opposite.
'If True' compiles to 'Jump False'.

Mnemonics

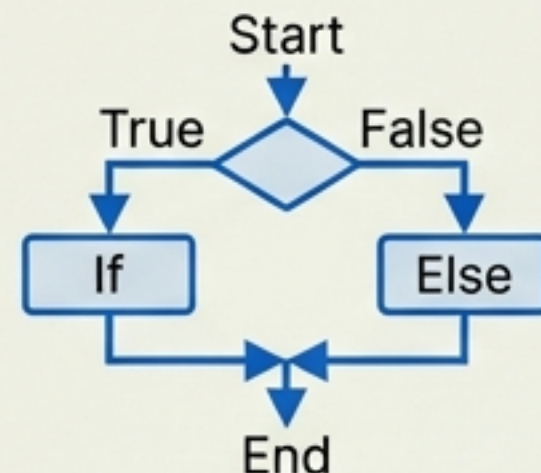


Signed = GL (Greater/Less).
Unsigned = AB (Above/Below).

Shape Recognition

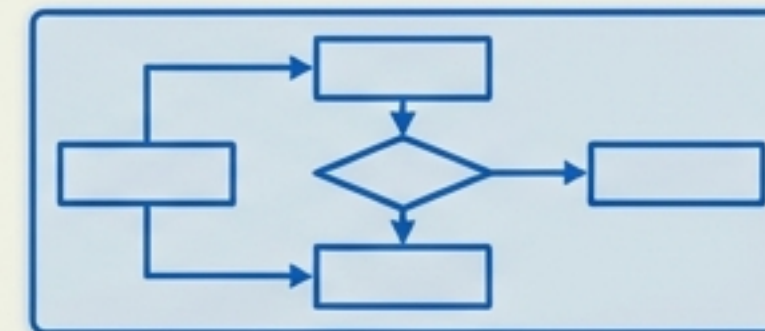


Loops = Upward Arrows.

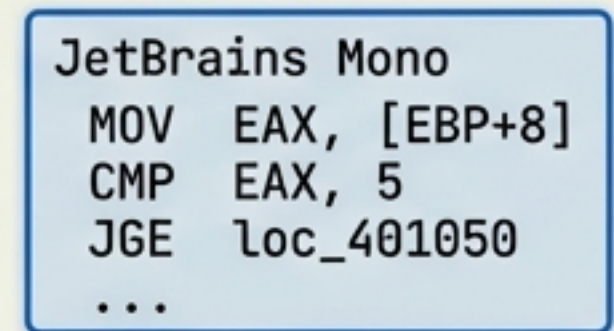


If-Else = Split & Join.

Views



Graph (Overview)



Text (Detail)

Graph for overview, Text for detail.